

EME: Education: Technology and Media Courses

Courses

EME 2040 Introduction to Emerging Technology

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Assists learners in developing skills and competencies which are essential to the integration of technology in various settings. Students will survey a wide variety of emerging technology systems and tools. They will also learn to use these tools in an organizational environment.

EME 2620 Digital Literacy in a Globally Connected World

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Digital technologies have become an influential part of our everyday life. For this reason, digital technology skills have become central across disciplines and are valued as an essential career competency. In this course, students learn how to access, evaluate, apply, participate, and interact within the educational and professional digital environments as they solve complex problems within a technology-rich world.

EME 3002 Intelligence and National Security

Col of Arts, Soc Sci and Human, Department of Criminal Justice

3 sh (may not be repeated for credit)

Students will develop an academic understanding of national security and the government agencies that are responsible for protecting the United States and its interests. Students will learn about the intelligence cycle, national security decision making, and the intelligence community and review case studies of intelligence in action. Students will also become familiar with analytic writing and intelligence analysis through case studies and weekly assignments of current national security news.

EME 3003 Open Source Intelligence

Col of Arts, Soc Sci and Human, Department of Criminal Justice

3 sh (may not be repeated for credit)

Provides students with an academic and practical understanding of Open Source Intelligence (OSINT) and its applications. Students will learn about Open Source Intelligence as a discipline, its place in the intelligence world, and OSINT planning and execution. Students will become familiar with OSINT acquisition and exploitation techniques by developing an understanding of available technological tools and capabilities.

EME 3233 Technology Integration Planning

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Provides students with the knowledge, skills, abilities, and attitudes necessary to implement instructional technology. Students will learn to identify the constraints and risks associated with instructional technology planning and implementation. Students will utilize software tools associated with the implementation of instructional technology.

EME 3312 Technology Supported Learning

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines the use of current and emerging technologies to facilitate learning. Topics covered will include distance learning, formal and informal technology based learning and mobile learning. Strategies for integrating technology in educational settings will be explored.

EME 3351 Introduction to Instructional and Performance Technology

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

The distinct purposes of instructional technology and human performance technology are explored in depth in this course. The foundations and evolution of each discipline serve to establish distinct definitions that will be investigated. The similarities and differences will be compared to include the historical basis, models, major tasks, and desired outcomes.

EME 3624 Training Needs Assessment

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines the role of training needs assessment in instructional design. Students will be introduced to techniques used to collect and analyze data to identify and clarify training needs. Prepares students to employ needs assessment techniques to determine who needs to learn what and why prior to engaging in the design and development of instructional materials.

EME 3905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 4001 HUMINT Operations

Col of Arts, Soc Sci and Human, Department of Criminal Justice

3 sh (may not be repeated for credit)

Students will learn the importance of human originated information, or HUMINT, in the context of law enforcement, military and intelligence operations. Students will learn about interview, interrogation and elicitation techniques that are employed within the law enforcement and national security communities. Students will be able to recognize and describe the difference between overt and clandestine source operations and when HUMINT should and should not be utilized in the pursuit of legal or national security priorities. Students will also be able to assess basic psychological indicators in the profiling of historic espionage cases and their impact on national security.

EME 4043 Instructional Technology Leadership

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will examine the role of the technology leader in effective integration, management and use of technology in a variety of settings, including education, training, military, public sector and non-profits. The course focuses on technology, information, and information literacy. Special attention is paid to the role of systems thinking in effective technology leadership.

EME 4083 Program Evaluation in Instructional Design and Technology

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will develop skills used in selecting the appropriate model for conducting various types of evaluations. A series of models will be reviewed and aligned with evaluation purposes and questions. Applying the appropriate evaluation model is critical to ensuring that interventions, programs, and projects are successful. Development of a comprehensive evaluation plan will provide students with the opportunity to align an evaluation model with data collection strategies and techniques for a specific evaluation purpose.

EME 4343 Multimedia Design and Development

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

The basic visual and typographical elements and technical aspects of multimedia design and development to support learning are the focus of this course. Students will apply instructional design strategies and principles of multimedia learning to the design and development of multimedia. Included are a selection of software applications and services, design principles, hands-on production, and discussion of issues and useful resources.

EME 4350 Human Performance Technology

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students are introduced to the field of Human Performance Technology (HPT). Through examination of the research, theories and models associated with HPT, students will be prepared to conduct comprehensive performance, gap and cause analyses in organizations, and identify training and non-training based solutions to resolve organizational performance concerns.

EME 4352 HPT Intervention Selection and Design

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Human Performance Technology (HPT) interventions are selected to resolve gaps in desired performance. The skills required to align interventions with the cause(s) of the problem are the focus of this course. Students will classify interventions using various models of Human Performance Technology and select potential interventions to resolve identified problems in human performance scenarios. Students will also develop a formal proposal to communicate recommendations to stakeholders.

EME 4474 Technical Intelligence Collection

Col of Arts, Soc Sci and Human, Department of Criminal Justice

3 sh (may not be repeated for credit)

Introduces students to intelligence disciplines (ELINT, SIGINT, MASINT, GEOINT) and intelligence organizations (NSA, NSA, NRO and DIA). Students will examine the history of these organizations, technologies used in each intelligence discipline, and common uses of each technology. The course focuses on improving analytical writing and research skills in the intelligence discipline.

EME 4673 Foundations of Instructional Design

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Introduces students to the field of instructional design, a systemic and systematic, research-based means of designing effective, efficient, learner focused instruction. Students will use the ADDIE process to design a lesson.

EME 4674 Development of Instructional Materials

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

The pedagogical, technical, and logistical aspects of instructional messages will provide the foundation for students to learn the fundamentals of instructional development in this course. Message design principles and individual preferences are considered as they relate to the development of instructional materials. Media and technology aspects relating to effective message delivery will be addressed and related to the logistical constraints of time and cost.

EME 4684 Instructional Design and Technology Capstone

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

The capstone is designed to enable students to demonstrate mastery of the Instructional Design and Technology knowledge, skills, and abilities developed during the academic program. Students will identify, propose, and complete a capstone project and develop an electronic portfolio highlighting their attainment of the program level learning outcomes. Permission is required.

EME 4905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 4940 Instructional Design Internship

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This internship course provides students with supervised, field-based experience in instructional design and technology settings. Students apply knowledge and skills gained in prior coursework to authentic professional contexts, engaging in activities such as instructional systems design, educational media development, technology integration, and performance improvement projects. Emphasis is placed on professional practice, reflective learning, and collaboration with site supervisors to gain practical insights into the roles and responsibilities of instructional design and technology professionals. Reports will be required on a regular basis with a final report and oral interview. Permission is required.

EME 5905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 6035 Foundations of AI Tools in Education

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This course introduces the fundamental concepts, mechanisms, and capabilities that underlie AI systems used in education. Students examine emerging technologies, including multimodal models, intelligent tutoring systems, and adaptive platforms, to understand how they shape learning environments. The course highlights strengths, limitations, and practical considerations when selecting and implementing AI tools.

EME 6036 Applied Project in AI-Supported Learning Environments

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This course guides students through the analysis, design, development, implementation, and evaluation of an AI-supported instructional product or learning activity. Students apply principles from curriculum theory, technology integration, and ethical practice to create a project aligned with authentic educational needs. The final project demonstrates the student's ability to plan, justify, and refine AI integration within a specific teaching or training context.

EME 6054 Foundations of Instructional Design and Performance Technology

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students investigate theoretical, historical, sociological, and philosophical perspectives and applications of instructional design and performance technology in education and training environments. Students develop the knowledge, skills, and abilities needed to integrate instructional and performance technology theories and processes into education and training settings. Students are introduced to the theoretical and philosophical foundations of the field, and they are empowered to develop a comprehensive definition of the field and a broad perspective of IDPT on educational and training settings.

EME 6409 Distance Learning Implementation

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Integrates theory and best practices to explore and develop skills for developing and implementing effective education and training environments delivered via distance learning media. Students will focus on the principles and practices that are research-based and result in quality distance learning experiences, and students will explore technologies available to support and distribute distance learning and the considerations unique to distance learning. The course focuses heavily on online environments, and it emphasizes application of the best practices by enabling students to develop and implement their own instructional lessons that are delivered via distance learning technologies.

EME 6414 Web-Based Instructional Tools

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Provides students with the knowledge and skills necessary to design and develop web-based instruction using current authoring software and services. The course integrates theory and application. Students will learn to critically examine the instructional capabilities of various technologies and identify instructional strategies that support integration. Students will design and develop multiple units of instruction that demonstrate their ability to author courseware.

EME 6415 Digital Video for Instruction

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Principles of instructional video design and development including designing for learning objectives, effective audio and lighting techniques, video recording, editing, and delivery will be taught. Students will explore the opportunities and technical challenges associated with web-based video as a communication medium. Practical application projects are an integral part of the learning experience as students explore all aspects of instructional video pre-production, production, and post-production.

EME 6427 Implementation of HPI Interventions

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Provides students with fundamental knowledge and skills related to the intervention selection, design, and implementation; and change management activities associated with the practice of Human Performance Improvement (HPI). Analyzes potential interventions to identify those most appropriate for identified root causes of performance gaps. Examines models of change management, the role of the change agent and the importance of developing and implementing effective change management plans to insure successful intervention implementation and institutionalization.

EME 6428 Evaluation of HPI Interventions

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will examine the theory and practice of evaluation models and processes as they relate to the formative, summative and confirmative evaluation of instructional and non-instructional Human Performance Improvement (HPI) interventions. Students will develop the knowledge, skills and abilities necessary to plan and conduct comprehensive evaluations based on current theories, models, and best practices.

EME 6429 Human Performance Improvement

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Provides students with fundamental knowledge and skills related to the performance, gap and cause analysis activities associated with the practice of Human Performance Improvement (HPI). Examines the importance of systems thinking in HPI and the theories and theorists of the field.

EME 6458 Distance Learning Policy and Planning

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines the history of distance learning and the principles, policies and issues related to the design, development, implementation and administration of distance learning courses and programs in various settings. Issues related to technology, teaching, learning, assessment and faculty and student preparation will be considered from both theoretical and practical perspectives.

EME 6607 Instructional Technology Integration Projects

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Provides students with the knowledge, skills, abilities, and attitudes necessary to implement instructional technology in educational and training environments. Students will learn to identify the constraints and risks associated with instructional technology planning and implementation and develop ways to manage these factors. Students will utilize software tools to manage the implementation of an instructional technology project.

EME 6609 Instructional Design Theory and Practice

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Theoretical underpinnings and practical applications for instructional design will ground the course. Students will examine the key components of the instructional system and the theoretical perspectives that inform the practice of instructional design. Students will apply a systems approach and a research-based model to the instructional design process to design a pedagogically sound instructional product.

EME 6626 Emerging and Innovative Technology Systems

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

New technology and approaches to teaching and learning evolve and revolutionize how professionals approach technology integration. Students will explore how innovation and emerging technologies can be integrated into instructional strategies to enhance teaching, learning, and performance.

EME 6678 Theoretical Foundations of Instructional Design

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will examine the key components of the instructional system and the theoretical perspectives that inform the practice of instructional design. The role of communication theories, learning theories, and instructional theories, and the overarching concept of alignment in instructional design will be explored.

EME 6905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 6940 Instructional Design Internship

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This graduate-level internship provides students with supervised professional experience in instructional design and technology settings. Students apply advanced theories, models, and practices to authentic projects such as instructional systems design, educational media development, technology integration, and performance improvement initiatives. Emphasis is placed on professional collaboration, ethical practice, project management, and reflective analysis to support the transition from graduate study to professional practice in instructional design and technology.

EME 6946 Instructional Design and Technology Capstone

School of Education, Department of Instructional Design and Tech

3 sh (may be repeated for up to 6 sh of credit)

Students critique the academic program, identifying their key learning outcomes, and the courses and specific instructional strategies that led to those outcomes. Students identify, propose, and complete a complex project, integrating knowledge, skills, and abilities developed in multiple classes to solve an instructional or performance related problem in a real organization. Permission is required.

EME 7067 Leadership in Performance Improvement

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will investigate the role of leadership in guiding performance improvement projects. Students will critique leadership techniques and mentoring strategies and learn to determine which approaches are most appropriate for given situations. Special focus will be given to systems thinking and its role in effectively leading performance improvement projects.

EME 7068 Analysis and Integration of Instructional Technologies

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students examine theoretical foundations, research-based strategies, and best practices used to analyze the value of current and emerging technologies in resolving organizational performance problems. Students will select and effectively integrate a broad range of technologies in diverse organizational settings.

EME 7353 Evaluation of Performance Improvement Interventions

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students examine strategies for planning, conducting, and leading evaluations of performance improvement interventions. Students will analyze evaluation types, approaches, and models, and develop the skills necessary to plan, conduct, and lead evaluation efforts aligned with the subject of the evaluation, available resources, and the constraints of the organization.

EME 7357 Intervention Selection, Design and Development Leadership

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will critically examine a wide range of potential performance improvement interventions to determine which solutions are best suited for various situations. Students will focus on aligning solutions with identified problems and organizational constraints and effectively communicating recommendations to stakeholders. Students will apply research, theory, and best practices to lead instructional and non-instructional intervention design and development projects.

EME 7365 Performance Improvement Theory and Research

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students examine the theoretical and conceptual foundations of the field of Performance Improvement through a comprehensive review and extensive examination of the literature of the field. Students will develop knowledge, skills, and abilities necessary to apply research and theory to practice to improve organizational performance.

EME 7366 Data Collection in Performance Improvement

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines data types and collection methods, sources, and instrumentation. Examines how, when, and why to collect quantitative and qualitative data for performance improvement projects. Critically investigates researcher bias, reflexivity, and positionality when conducting data collection for performance improvement research.

EME 7367 Data Analysis in Performance Improvement

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines the various types of data analysis consistent with performance improvement. Appropriate qualitative and quantitative data analysis procedures and alignment with research questions and study purposes are addressed in this course.

EME 7613 Principles of Instructional Systems Design

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students will examine the use of instructional systems design models to create instruction that is appropriate from a pedagogical and practical viewpoint. Theories and models to support the design of instruction for use in a variety of instructional formats will be emphasized. Focus areas will include analysis, instructional goals and objectives, assessment, instructional strategies and the role of formative evaluation in instructional design. Students will apply theories and best practices to design a pedagogically sound instructional product.

EME 7618 Fundamentals of Practitioner-based Research

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This course provides an introduction to the design of improvement-focused research in the field of instructional and performance technology. Students will examine principles of applied and action research, including how cycles of inquiry connect to the practice of identifying, framing, and addressing Problems of Practice in organizational contexts. Through critical analysis of the research literature, emphasis on methodological rigor, and reflection on responsible research practice, this course will guide the development of research skills that will frame the Dissertation in Practice.

EME 7692 Leading Performance Improvement Projects

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

This course focuses on the development of the skills necessary to effectively lead instructional and non-instructional performance improvement projects. Critical skills included in the course include leading projects and change initiatives, facilitating organizational communication, responsible standards of behavior, and social impact. Coursework will also provide students with opportunities to continue to develop skills in critical thinking, scholarly research, and professional writing.

EME 7695 Design of Non-Instructional Performance Solutions

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Examines the research, theories, and best practices that guide the design of non-instructional interventions/solutions to individual and organizational performance improvement opportunities in the workplace. Students will consider constraints, risks, and required resources for a variety of interventions/solutions, resulting in the development of one or more process models that may be used by practitioners to guide design activities.

EME 7905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 8608 IPT Foundations, Issues and Trends

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Students examine the history and evolution of the field of instructional and performance technology and its three major areas of emphasis; instructional design, instructional technology, and performance technology. Students will also analyze current issues and trends influencing the field and their impact on research and practice. This course focuses heavily on research, critical thinking, and communication skills.

EME 8693 Analysis and Dissemination of IPT Research

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Critically examines the types of written reports and oral presentations practitioner-scholars use to communicate findings and recommendations to stakeholders. Students will learn how to tailor written and oral communications to meet the culture of the organization and the information needs of the audience.

EME 8905 Directed Study

School of Education, Department of Instructional Design and Tech

1-12 sh (may be repeated indefinitely for credit)

EME 8980 Dissertation

School of Education, Department of Instructional Design and Tech

1-6 sh (may be repeated for up to 18 sh of credit)

Major individual research in an area of significant educational interest; designed specifically for candidates in the Ed.D. Curriculum and Instruction, Instructional Technology program. The dissertation reflects intensive educational research produced by the student and collaboratively developed with the student's graduate committee. Graded on a satisfactory / unsatisfactory basis only. Admission to candidacy, completion of all other doctoral program requirements and permission is required.

EME 8981 Doctoral Seminar: Dissertation in Practice- Phase 1

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Prerequisite: EME 8693

This is the first of a three-sequence doctoral seminar series designed to guide students through the early stages of their dissertation research. During this course, students will refine their problem of practice, develop appropriate research questions, and articulate the significance of their issue within their field. Emphasis will be placed on developing scholarly writing skills, improving synthesis of literature, and mastering APA style. Students will be paired with a dissertation chair, identify their committee members, and refine their Problem of Practice Statement or Statement of Opportunity.

EME 8982 Doctoral Seminar: Dissertation in Practice- Phase 2

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Prerequisite: EME 8981

This is the second course in a three-sequence doctoral seminar series, designed to support students in advancing their dissertation research. During this seminar, students will work closely with their instructor, dissertation chair, and committee to finalize and defend Deliverable 1: A Detailed Analysis Plan for investigating their problem of practice. Emphasis will be placed on selecting appropriate research methodologies, designing a comprehensive data collection and analysis strategy, and ensuring alignment between research questions and methods. By the end of the course, students will be prepared to formally present and defend their analysis plan to their dissertation committee.

EME 8983 Doctoral Seminar: Dissertation in Practice - Phase 3

School of Education, Department of Instructional Design and Tech

3 sh (may not be repeated for credit)

Prerequisite: EME 8982

This is the third course in a three-sequence doctoral seminar series, guiding students through the final preparations to implement their quality improvement interventions. During this seminar, students will work closely with their instructor, dissertation chair, and committee to finalize and defend Deliverable 2: A Comprehensive Intervention Selection, Design, and Development Plan for addressing their problem of practice. The course will focus on selecting evidence-based interventions, designing implementation strategies, and establishing evaluation metrics. By the end of the course, students will formally present and defend their intervention plan to their dissertation committee.

EME 8984 Dissertation in Practice: Continuation

School of Education, Department of Instructional Design and Tech

1 sh (may be repeated for up to 6 sh of credit)

Prerequisite: EME 8983

Facilitates continuous enrollment for students who have completed all required coursework without successfully defending the dissertation-in-practice and/or meeting other graduation requirements. Students enrolled in this course will work with their dissertation-in-practice committee chair to complete all remaining requirements to successfully graduate.